

Evaluate a Corrosion Claim

Complete every numbered item and submit this PDF.

Student name:

Scenario

A coating company claims it stops all corrosion permanently. After 30 days, uncoated iron gained 0.42 g and developed a thick reddish-brown coating. Coated iron gained 0.04 g and developed two small rust spots where scratched. A diagram shows oxygen and water reaching iron through breaks in the coating.

1. MODEL - what the diagram represents and one limitation

2. SCIENTIFIC LAW - a consistent pattern, not an explanation

3. THEORY-LEVEL EXPLANATION - why scratches affect corrosion

4. TESTABLE HYPOTHESIS - follow-up 60-day comparison

Decision, Evidence, and Defense

Use both mass measurements and the visible observations.

5. CLAIM DECISION

Is the company claim fully supported, partly supported, or not supported?

6A. REQUIRED EVIDENCE AND TRADEOFF ORGANIZER

Uncoated sample - mass and appearance

Coated sample - mass and appearance

Model limitation

Tradeoff

6B. SIX- TO EIGHT-SENTENCE DEFENSE

FINAL CHECK / CORRECTION RECORD

Confirm all six items are complete. Record any correction without erasing the original response.